Making glass from British sand

A practical route from a qualified British silica deposit to a consistent, saleable glass product.

Yes: Britain has silica sands suitable for glassmaking. The essential first step is to qualify a specific sand grade and finished-glass application, then prove the batch in controlled melting trials.

The core transformation	What happens
British silica sand	Provides the silica network; purity, iron and grain size influence the result.
Soda ash + stabilisers	Flux and calcium/magnesium-bearing ingredients help create a workable, durable soda-lime glass.
Compatible recycled glass	Cullet contributes an already-formed glass composition and can reduce virgin-material demand.
Controlled heat and cooling	Melting/refining, forming and annealing produce the final article; inspection confirms fitness for use.

Inside: British sand candidates, laboratory tests, an illustrative 100 kg glass batch, ten production steps, equipment, quality control, environmental duties and a staged start-up plan.

Scope: lead-free soda-lime-silica glass, with decorative cast/pressed products as the recommended first market. Solar cover glass, food containers, structural glazing and pharmaceutical glass require additional qualification.

Public-source research and original engineering calculations. This is a development guide, not a validated furnace recipe, guaranteed supplier specification or investment quotation.

The practical start-up route

Begin with the product, then the sand

A credible first venture would develop a small range of premium decorative cast or pressed objects using a documented British-sand batch. Start with laboratory/contract melting, prove appearance and durability, then assess an owned small furnace. This is a recommended business sequence for an unspecified start-up, not evidence of a guaranteed market or margin.

Soda-lime glass combines silica sand with soda ash and stabilising ingredients. Soda ash makes melting more practical; calcium-bearing material improves durability. Ordinary glass is an amorphous solid, with a disordered atomic structure rather than the long-range crystal order of quartz. [7,8]

Three ways to enter

Route	What you actually do	Key implication
Develop a British-sand batch with a contract melter	Specify raw materials and composition; commission trials/limited production; own design and quality requirements.	Best initial fit for proving the sand-to-product proposition.
Operate a small melting/forming shop	Buy qualified sand/minerals/cullet; melt, form, anneal and inspect in-house.	Requires specialist staff, utilities, furnace/refractory knowledge and compliance.
Process bought-in glass	Cut, polish, decorate or reshape sheet/cullet from another producer.	Lower upstream scope, but British manufacture does not prove British sand origin.

A new float-glass line is a fundamentally different project from a small glass shop. Float production continuously melts, refines, forms on molten tin and anneals a ribbon. Pilkington describes a long, tightly integrated process; access to sand alone is insufficient to justify that scale. [9]

The first commercial decision should be a customer brief: form, colour, dimensions, finish, volume, price and intended use. “Made with British sand” can support a provenance story, but does not replace measured quality, safe use or viable unit economics.

Where British glass sand could come from

Candidate contacts below are grounded in published producer and geological information. None is a confirmed supply agreement, stock check, delivered quotation or guarantee that a particular grade meets this venture's specification.

Candidate	Published evidence	What to request
Lochaline, Morvern, Scotland	Producer parent describes low-iron LQS85 and LQS500 for glass applications. BGS also documents commercial grades. [3,6]	Current grade/Fe2O3 limits, full chemistry, particle distribution, available volume, origin and delivered terms.
Sibelco UK / Talke, Staffordshire	Sibelco lists Talke as supplying silica mainly for ceramics and glass. A processing/supply site is not by itself proof of extraction origin. [4]	British quarry of origin for the proposed grade, lot certificate, contamination control and glass-use evidence.
Bathgate / Arclid, Cheshire	Bathgate's refined-sand sheet identifies Arclid Quarry and publishes indicative particle grading. It does not by itself establish an optical-glass grade. [5]	Full oxide/trace analysis, iron specification, variation across lots and a sample for melting trials.

Use geology as a screening tool

BGS's 2020 silica-sand factsheet describes Britain's restricted industrial-sand resources. Its 2025 market factsheet emphasises the specific chemical and physical characteristics demanded by end users and the limited sources able to supply them. A mapped deposit is not a permitted, available or commercially suitable feedstock. [1,2]

For a low-iron clear product, Lochaline is a well-supported first enquiry. For tinted decorative glass, a different qualified grade may be more economical. Colour tolerance must be agreed in the finished article, including its thickness, rather than chosen solely by the colour of dry sand.

Avoid a common error: one BGS waste-resource sample from Lochaline analyses hydrosizer fines, not a shipment of a named commercial glass-sand grade. Its assay must not be presented as the commercial product certificate. [6]

How to qualify the sand

Measurement	Why it matters	Proposed procurement evidence
SiO2 and major oxides	Glass composition and batch correction; clay/feldspar/carbonate contributions.	Laboratory XRF or appropriate validated chemical method; dry-basis report.
Iron reported as Fe2O3	Colour and optical absorption, especially for thick/extra-clear products.	Agreed upper limit, analytical method and lot variability. Distinguish Fe from Fe2O3.
Other colourants / minerals	Chromium, titanium, manganese and heavy minerals can affect colour/defects.	Trace analysis plus mineralogical examination where relevant.
Grain-size distribution	Dissolution, segregation, dust and melting behaviour.	Sieve/particle-size results and agreed out-of-size fractions.
Moisture / loss on ignition	Wet weighing and gas generation; consistency of batch calculations.	Separate moisture from other ignition losses; define the reporting basis.
Refractory grains / contamination	Unmelted inclusions and product rejects; transport contamination.	Microscopy/mineralogy, clean handling records and controlled melting trial.

Practical sampling sequence

Request representative samples from several lots, with a sampling statement and certificates. Retain sealed reference samples. Use an independent laboratory to verify the proposed acceptance methods; do not set a limit lower than the analytical method can reliably distinguish.

Run a known reference batch and the candidate sand under comparable melting conditions. Compare composition, colour/transmission, inclusions, bubbles, viscosity-related working behaviour and annealed quality. GTS offers raw-material melting trials and composition/property development; ask for an agreed scope and deliverables. [11]

Processing has limits

Washing, classification, attrition and suitable separation may improve a feedstock, but cannot be assumed to remove every mineral or lattice-bound impurity. Test beneficiation on the actual deposit and account for water, tailings, yield and permits. For a first venture, purchasing a proven processed grade is usually a more manageable route than creating a quarry and purification business at the same time.

What goes into soda-lime glass?

Ingredient / contribution	Function and design consideration
Silica sand → SiO2	Forms the main glass network. Sand purity and particle size must suit the melt and optical requirement.
Soda ash → Na2O equivalent	Acts as a flux/network modifier. The carbonate releases CO2 during the reaction process.
Limestone → CaO equivalent	Adds calcium for property control and durability; also generates carbonate-process CO2.
Dolomite → CaO + MgO equivalents	Supplies both calcium and magnesium. Account for both oxides instead of treating it as pure limestone.
Alumina-bearing materials	Adjust properties and durability. Feldspar adds several oxides; high-purity alumina is used only as an accounting simplification in the example.
Compatible cullet	Already-formed recycled glass supplies several oxides together. Its composition, colour and contamination change the required virgin batch.
Minor refining / colour ingredients	Product-specific. Redox, emissions, refractory compatibility and health/environment duties require review.

Pilkington lists sand, soda ash, dolomite, limestone, saltcake and cullet among flat-glass inputs. The exact batch differs by product and equipment. British Glass also distinguishes raw batch mass from finished-glass composition because carbonate ingredients lose gas during manufacture. [8,19]

A proposed calculation basis

The next page uses a notional final composition of **72% SiO2, 14% Na2O, 10% CaO, 3% MgO and 1% Al2O3** by mass. This is an illustrative soda-lime-silica design basis, not a qualified recipe for a food container, solar panel, window or thermal-shock product.

Oxide accounting expresses what each raw material contributes to the final glass; it does not mean that free sodium oxide and calcium oxide are separately added. A real industrial calculation uses the complete assays of every raw material and cullet, with losses and contamination measured from trial melts.

Worked batch: 100 kg of theoretical glass

Assumptions: pure dry silica, anhydrous soda ash, pure dolomite, pure limestone and alumina; complete retention of target oxides; no dust, volatilisation or handling losses. All numbers below are original stoichiometric calculations, rounded for explanation. They are not furnace charging instructions.

Raw ingredient	No cullet: kg	With 30 kg matching cullet: kg
Silica sand	72.00	50.40
Soda ash, Na2CO3	23.94	16.76
Dolomite, CaMg(CO3)2	13.73	9.61
Limestone, CaCO3	10.40	7.28
Alumina, Al2O3	1.00	0.70
Cullet, same oxide composition	0.00	30.00
Total theoretical dry charge	121.06	114.75
Finished glass, theoretical	100.00	100.00
Carbonate CO2, theoretical	21.06	14.75

The no-cullet calculation uses oxide mass fractions: soda ash contributes about 58.48% Na2O; dolomite about 21.86% MgO and 30.41% CaO; limestone about 56.03% CaO. Supply the 3 kg MgO through dolomite, deduct its calcium contribution, then supply the remaining CaO through limestone.

Correct the actual weighing basis

With 30% of the theoretical final glass supplied by identical-composition cullet, each virgin input falls to 70% of its no-cullet amount. If the sand has 4% moisture by as-received mass, weighing 50.40 kg dry sand requires **50.40 / 0.96 = 52.50 kg** as received. Impurity corrections require the full assay, not this moisture adjustment alone.

Iron example: 50.40 kg sand at an assumed 85 ppm Fe2O3 contributes 4.284 g Fe2O3, or 42.84 ppm in 100 kg theoretical glass, before iron from cullet, other ingredients, tools and refractories. The 85 ppm value is an assumption for arithmetic, not a certificate for a current supplier lot.

Steps 1-3 / Product, materials and batch

01 / Freeze the article and glass specification

Choose one product family and intended use. Define dimensions, colour/optical acceptance, finish, thermal/mechanical exposure, durability and customer tests. Select the glass family and forming route with a glass technologist. Decorative paperweights or display objects can provide a simpler first brief than pressure bottles or safety glazing.

Check before proceeding: Approved product drawing, intended-use statement, glass-property targets and acceptance plan.

02 / Purchase and release the raw materials

Obtain a qualified British sand grade plus the selected soda ash, stabilisers, alumina-bearing material and cullet. Inspect and test against the procurement plan. Verify origin separately from shipping address. Segregate cullet by compatible glass family/colour and remove unacceptable ceramic, stone, metal and other contaminants.

Check before proceeding: Origin documents, supplier/lot IDs, assay and moisture data, cullet acceptance and retained samples.

03 / Calculate, weigh and mix the batch

Solve the oxide balance using the real analyses. Apply moisture and loss corrections. Weigh with calibrated equipment in the engineered dust-controlled system; mix to the validated uniformity. Label and segregate the batch to prevent wrong additions or cross-contamination. Use a qualified premix where it is safer or more consistent.

Check before proceeding: Signed batch sheet, scale IDs, actual weights, recipe revision, mixing record and material genealogy.

Hold point: do not proceed to hot trials when sand/cullet identity is uncertain or the mass balance is unexplained. The example on the previous page is only the starting arithmetic; a specialist must qualify the real batch and its furnace compatibility.

Provenance record

Record the extraction site and the processed grade for the sand, the country/site of each other raw material where known, the cullet stream and where glass was melted and formed. If several sources are blended, retain the blend proportions and lot links.

Steps 4-6 / Melt, refine and condition

04 / Prepare the approved melting system

Use a furnace and refractory/crucible system qualified for the composition, temperature, throughput and forming route. Follow the supplier’s inspection, dry-out, heat-up and charging procedure. Verify extraction, sensors, interlocks, containment and emergency arrangements before introducing material.

Check before proceeding: Equipment/refractory revision, commissioning records, temperature calibration, approved operating limits and hot-work release.

05 / Melt the batch under controlled conditions

Charge through the designed system using the validated rate and material condition. Monitor the actual melt and furnace behaviour, not only the controller setpoint. Track batch dissolution, foaming, temperature distribution, emissions and refractory interaction. Prevent uncontrolled wet contamination or incompatible additions.

Check before proceeding: Batch ID, charge history, energy/fuel use, temperatures, operator observations and hold/reject events.

06 / Refine, homogenise and reach forming condition

Use the developed time-temperature/redox/refining strategy to reduce bubbles and composition gradients. Then condition the glass to the viscosity and uniformity required by the forming method. Sampling and microscopy help distinguish persistent inclusions from bubbles or refractory defects.

Check before proceeding: Glass sample results, bubbles/inclusions, colour/composition, working behaviour and approved forming release.

Temperature context: NSG describes glass synthesis at roughly 1,550°C; Verallia describes a container-glass process at 1,550°C; Pilkington’s float account includes furnace conditions around 1,700°C. These are descriptions of different systems, not interchangeable operating setpoints or a universal 24-hour recipe. [18,10,9]

Set a trial schedule from composition, particle size, batch amount, refractory, furnace heat transfer and required quality. Scale-up changes heat flow and residence-time distribution; a successful crucible sample does not establish a production tank’s performance.

Steps 7-10 / Form, anneal and release

07 / Form the article

Cast or press the glass using tooling, handling and mould conditions qualified for the design. Control temperature/viscosity, fill, release and dimensional variation. For blown containers or a flat ribbon, use the corresponding specialised process rather than adapting a small casting procedure.

Check before proceeding: Tooling ID, forming conditions, part weight/dimensions, defects and operator/lot traceability.

08 / Anneal through a validated cooling schedule

Use the composition-specific annealing range and a schedule suitable for the article’s thickest section and geometry. Control temperature uniformity, hold and cooling through the critical range. Record the actual cycle and evaluate residual stress; large/thick castings can need very different cycles from thin articles.

Check before proceeding: Calibrated annealer data, loading plan, cycle revision and residual-stress assessment.

09 / Finish and perform any further treatment

Cut/grind/polish as specified with appropriate dust/wet-process controls. Inspect edge quality. Apply decoration or coatings only through a qualified process. Toughening, lamination and chemical strengthening are separate operations with additional design and testing requirements.

Check before proceeding: Finishing/inspection record; treatment qualification where applicable; no unexplained edge damage.

10 / Inspect, package and release

Check appearance, geometry, composition/property evidence and application-specific tests. Segregate rejected glass into a traceable compatible return stream. Package to prevent surface damage and impact; provide intended-use, care and batch information. Authorise release only after deviations are resolved.

Check before proceeding: Quality sign-off, test results, label/claim review and retained reference article.

Annealing relieves unwanted internal stress; it does not automatically create toughened safety glass. GTS describes residual-stress inspection using polarised light, alongside dimensional, thermal-shock and mechanical quality assessments. [21]

Equipment and forming-route choices

Route	What it requires	Start-up implication
Pot / crucible melting + casting or pressing	Batch preparation, compatible furnace/crucibles, forming tools, annealer and finishing/test capability.	Useful for development or suitable specialist volumes; energy and labour per kg may be high.
Continuous small tank / day tank	Engineered charging, melt/refining, controlled withdrawal, forming and annealing with coordinated utilities.	Evaluate only after repeat demand, process stability and furnace-life economics are credible.
Automated container production	Consistent melt delivery, gob/feeder system, forming machines/moulds, lehr, inspection and packing.	Specialist project; product design, mould cost and line utilisation dominate decisions.
Float sheet production	Continuous large furnace, tin-bath system, atmosphere control, annealing ribbon and cutting.	A major integrated industrial project; not the default launch route.
Rolled / patterned sheet	Controlled melt delivery, rollers/forming, annealing and surface/geometry inspection.	Requires its own pilot and market case; not equivalent to optical-quality float.

Specify equipment using the real process

Request the supplier’s supported glass compositions, temperature/throughput envelope, power/fuel needs, refractory life assumptions, cooling/extraction requirements, floor/loading requirements, controls, data export and service plan. Confirm how start-up/shutdown, power failure and maintenance are handled.

Use development facilities first

GTS advertises small-scale melting and forming trials. Glass Futures reported first rolled glass from its experimental 30-tonne-per-day furnace in September 2025. The latter is an industrial R&D; resource, not evidence that a small firm can book any batch or forming route on demand. Confirm compatibility, access, IP, sample output and cost directly. [11,12]

Quality tests and defect investigation

Issue / property	Investigation and proposed evidence
Green/yellow tint or poor transmission	Whole-batch iron/colourant balance, thickness, redox, cullet and refractory contamination; spectral transmission/colour against a reference.
Stones / unmelted inclusions	Sand grain/mineral distribution, batch dissolution, refractory spalling and contamination; microscopy and composition.
Seeds / bubbles	Batch gases, melt/refining conditions, contamination and refractory reactions; quantified inclusion criteria.
Cords / striae	Composition or thermal nonuniformity and material segregation; optical/chemical assessment appropriate to the product.
Cracking after cooling	Annealing history, thickness gradients, residual stress, tool/edge damage and mechanical handling.
Surface defects / devitrification	Composition, contact/tooling, reheating/cooling history and crystallisation; surface and phase analysis as needed.
Weak chemical or thermal performance	Actual composition and intended exposure; validated durability, thermal-shock and mechanical testing.

Define acceptance before trial production

Specify what is measured, at what sample condition, with which method and frequency. For optical glass, define thickness, spectral range and measurement geometry; “clear” is not a complete technical specification. For a decorative object, visible defect limits and residual stress still need an agreed basis.

GTS offers visual/design examinations, annealing assessment, thickness/weight checks and thermal-shock/impact work. Container pressure strength and other end-use tests require their own protocols. Select the relevant test package instead of claiming that one general glass test establishes all applications. [21,22]

Keep the production history

Link quarry/grade and raw-material lots to recipe, melt, furnace/refractory campaign, forming tool, annealing cycle, finished batch and customer shipment. Investigate defects using retained sand, glass coupons and ordinary production articles, rather than only the best demonstration piece.

Site safety and environmental duties

Engineer dust and hot-process controls

Dry silica handling can expose workers to respirable crystalline silica. HSE lists a GB exposure limit of 0.1 mg/m³ as an 8-hour time-weighted average and requires COSHH risk control; this is not a design target to operate up to. Use enclosed transfer, suitable extraction, appropriate cleaning and verified exposure control, with RPE and health surveillance where indicated. [13,14]

Provide furnace/hot-glass containment, guarded machinery, lifting/handling aids, suitable PPE, ventilation and trained response to spills, power failure or cooling-system faults. Keep material moisture within the qualified charging method; uncontrolled water contact with very hot material can cause violent steam generation. HSE also identifies handling, moving objects and machinery as important glass-sector injury causes. [23]

Check the permit route before signing the site

In England and Wales, Schedule 1 Section 3.3 distinguishes glass activities by type and capacity. It includes A(2) glass manufacture above 20 tonnes/day, with separate A(1) provision for glass fibre, and Part B categories including 5,000 tonnes/year capacity or lead-related glass manufacture, subject to the detailed wording. Confirm classification against the complete proposed installation. [15]

Do not interpret being below one threshold as exemption from every duty. Ancillary combustion, emissions, waste/cullet status, water discharge, planning, building/fire requirements and other operations may matter. Ask the relevant regulator to confirm the scope. The England/Wales route is not a substitute for checking SEPA in Scotland or the Northern Ireland authorities. [16]

Qualify the intended use

Food/drink contact, pressure containers, candle vessels, safety/structural glazing, solar covers and medical uses need specific requirements and evidence. UK guidance on food packaging highlights suitability for food use; do not issue a blanket food-safe claim based only on the absence of added lead. [17]

Economics: cost the saleable article

This is a hypothetical small-melting scenario in GBP, not a 2026 quotation or a forecast. Its purpose is to show the effect of utilisation, yield and actual energy consumption. A specialist furnace business case needs equipment, utility and customer quotations.

Illustrative input	Assumption
Nominal finished-melt capability	100 tonnes/year
Operating utilisation	70% → 70 tonnes gross glass/year
Electricity intensity / price	2,500 kWh per gross tonne × £0.20/kWh = £500/tonne
Other variable cost	£350 per gross tonne; hypothetical combined materials/consumables
Annual fixed cost	£120,000 illustrative aggregate; financing/selling costs excluded
Saleable yield cases	70%, 85%, 95%; no return-cullet credit modelled

Saleable yield	Saleable tonnes/year	Modelled manufacturing £/kg
70%	49.0	£3.66
85%	59.5	£3.02
95%	66.5	£2.70

Calculation: annual modelled cost = 70 × (£500 + £350) + £120,000 = **£179,500**. Divide by saleable kg. A 0.5 kg article at the 85% case carries about £1.51 of this modelled manufacturing cost before any excluded cost, article-specific work or commercial margin.

Replace every assumption with measured evidence

Include furnace standby, heat-up, annealing, forming and finishing energy; staff, tooling/moulds, refractory replacement, maintenance, testing, packaging, freight, waste, insurance, rent, finance and working capital. Electricity billed to a small shop is not comparable to energy figures for a continuously operated large tank.

Price the British sand grade delivered to the site, including moisture, transport, minimum order and loss/yield effects. Low-iron material is worth a premium only if the finished product needs and preserves its optical benefit.

British provenance and lower-carbon glass

Make the origin claim precise

“Melted and formed in Britain using sand extracted at [named British site]” is a more specific claim than “100% British glass” when other minerals or cullet have different or uncertain origins. Retain invoices, grade/lot records and extraction provenance. A British distributor address does not establish British extraction.

Cullet can support recycling and reduce virgin batch demand, but its original sand origin may be unknown. State recycled content on an explicit mass basis and explain the relevant boundary. Do not imply that the recycled fraction was made from newly purchased British sand.

Separate process and energy emissions

The worked mass balance produces about 21.06 kg of carbonate-process CO2 per theoretical 100 kg glass without cullet, or 14.75 kg with 30% identical-composition cullet. These figures exclude fuel, electricity, quarrying, transport and all other life-cycle effects. Electric melting does not remove the carbonate reaction emissions.

Measure full-site energy per saleable kg and carbon on a declared boundary. Include furnace/refractory life, material preparation, transport, finishing and rejects as appropriate. Renewable electricity, recycled content and local sourcing are separate facts; none by itself proves a zero-carbon or environmentally superior product.

Protect the low-iron objective

If clear/high-transmission glass is the business case, control iron and other colourants in cullet, stabilisers, handling tools and refractories. Merely choosing a low-iron sand cannot guarantee low-iron finished glass. Quantify transmission and colour at the final thickness.

NSG’s materials reporting describes raw-material use and identifies an interest in Lochaline Quartz Sand, supporting the industrial relevance of the source. That relationship is not an offer to supply this venture. [18]

Apply the CMA Green Claims Code when making environmental claims: use current evidence, clear scope and fair comparisons, and avoid a selective claim that misleads about the whole product. [20]

A staged route from samples to a business

Phase	Work	Proposed exit gate
0-3 months / define	Choose one product; engage a glass technologist; obtain sand samples and chemistry; agree customer tests.	A credible product brief and technically plausible material supply.
3-6 months / prove	Reference and candidate melts; compare cullet levels, colour, inclusions, working and annealing; make prototype articles.	Repeatable material/property results and customer feedback.
6-12 months / qualify	Pilot lots; tooling/finishing trials; residual stress and use-specific tests; compliance/site/utility review.	A controlled recipe/process, cost model and qualified product.
12-24+ months / scale	Contract production or commissioned small furnace; supply agreements; staff and quality system; monitored customer releases.	Saleable yield, durable demand and finances support expansion.

Equipment procurement gate

Before purchasing a furnace, obtain acceptance tests using the actual batch and intended article. Agree output, temperature uniformity, energy measurement boundary, refractory assumptions, utility interfaces, emissions, failure modes, training, spare parts and service response. Include any restrictions on third-party raw materials or cullet.

Capital planning

No reliable capital figure follows from the words “glass start-up”. A contract-melted design business, a staffed hot shop and a continuous container or float factory are different projects. Obtain separate quotations for technical trials, site work, utilities, furnace, forming equipment, annealing, testing, abatement and commissioning.

Reserve working capital for raw material, tooling, furnace ramp, trial rejects, inventory and customer payment delays. Model loss of a sand grade, higher energy prices, long furnace downtime and a customer qualification delay. Do not treat grants or an unsigned customer forecast as secured finance.

The first 90 days and supplier enquiry

Days 1-30: establish the facts

Interview prospective customers and choose one article/application. Send the specification below to at least two credible sand sources and an independent laboratory/development partner. Gather representative samples, extraction provenance, delivered quotes and technical records. Prepare the site/utility and regulatory questions before committing to a location.

Days 31-60: melt and compare

Run an agreed reference and candidate trial matrix. Hold composition and process variables sufficiently controlled to identify causes. Use the real sand/cullet assays in batch calculation; retain samples and full melt/annealing histories. Measure colour, inclusions, dimensions, residual stress and product-relevant durability.

Days 61-90: make an investment decision

Review customer acceptance, trial reproducibility, saleable yield and complete cost. Compare contract manufacture against an owned furnace. Produce a model-specific process brief, quality plan, supply risks, environmental route and quote-backed budget before ordering equipment.

Ask the sand supplier for	Required answer / attachment
Identity and origin	Grade, extraction site, processing site, country and lot traceability.
Chemistry	SiO2, Fe2O3 and other major/trace oxides; method, dry basis and guaranteed limits.
Physical data	Particle distribution, fines/oversize, moisture, loss on ignition and consistency across lots.
Glass-use evidence	Comparable application, reference samples and any independent trial data.
Commercial offer	Representative samples, quantity, minimum order, packaging, lead time and delivered price.
Continuity / changes	Available supply, notifications of material/process/source changes and backup options.

Decision standard: British provenance, a qualified composition, repeatable annealed product quality, a customer willing to pay and a viable manufacturing cost. Each should have evidence attached to the investment paper.

Sources and evidence notes

Research cut-off: 4 November 2026. Targeted public-source review of geological, producer, technical-service and regulatory material. No supplier lot, extraction site, furnace or product was independently sampled or tested. Calculations, operating recommendations and schedules are original illustrative synthesis.

[1] British Geological Survey. Mineral planning factsheet: silica sand, 2020. Open source
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[2] British Geological Survey. Industrial sand future markets, November 2025. Open source
Application-specific sand characteristics and constrained sources.

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Arclid source and indicative grading; not an optical-glass suitability certificate.

[6] British Geological Survey. UK Mineral Waste Factsheets, OR/24/040. Open source
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Manufacturer description of ingredients.

[9] Pilkington. The Float Process Step by Step. Open source
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[10] Verallia. The stages of glass manufacturing. Open source
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[11] Glass Technology Services. Glass Design & Development. Open source
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[12] Glass Futures. First Glass Successfully Rolled from Experimental Furnace, 30 September 2025. Open source
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Task-related exposure control guidance; select controls for actual operations.

[15] UK legislation. Environmental Permitting (England and Wales) Regulations 2016, Schedule 1 Section 3.3. Open source
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[17] GOV.UK. Food labelling and packaging: overview. Open source
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[18] NSG Group. Materials. Open source
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[19] British Glass. Recycled content in glass packaging, 2019. Open source
Batch-versus-final-composition distinction and recycled-material context.

[20] CMA / GOV.UK. Green Claims Code checklist. Open source
Evidence and clarity for environmental claims.

[21] Glass Technology Services. Glass Quality Assessment. Open source
Residual stress, dimensions, thermal shock and impact assessment capability.

[22] Glass Technology Services. Glass Fit For Purpose Testing. Open source
Application-specific strength and thermal performance tests.

[23] HSE. Glass and glazing. Open source
Industry injury and occupational-health context.

Before a production recipe is released

Obtain representative assays, qualify raw-material and cullet compatibility, trial the full composition, establish melt/forming/annealing windows, validate the intended-use tests and confirm site duties. The 100 kg batch is pure-material oxide accounting; real raw materials require a complete assay-based balance and qualified operating procedure.